

Self-Discrepancy Conditional Independence Test

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Abstract

Tests of conditional independence (CI) of random variables play an important role in machine learning and causal inference. Of particular interest are kernel-based CI tests which allow us to test for independence among random variables with complex distribution functions. The efficacy of a CI test is measured in terms of its power and its calibratedness. We show empirically that increasing the number of bootstraps in the Kernel CI Permutation Test (KCIPT) can increase rejection rates while degrading calibration. To address this limitation, we propose a novel CI test, called Self-Discrepancy Conditional Independence Test (SDCIT). SDCIT uses a statistic derived from the unbiased estimator of squared maximum mean discrepancy (MMD), applied to a sample and its permuted counterpart with selected terms omitted. This modification is not in general an unbiased estimator of a population squared MMD. The original experiments reported that SDCIT is, relative to the other methods: (i) competitive in terms of its power and calibratedness, outperforming other methods in the high-dimensional conditioning settings studied; (ii) robust across the tested Gaussian-kernel bandwidths; and (iii) competitive in run time on these benchmarks. These results have not been recomputed after the implementation corrections.

can be expressed as $P_{xz}P_{y|z}$. Tests of conditional independence (CI) play a central role in statistics (Dawid, 1979), machine learning including dimensionality reduction (Fukumizu et al., 2004, 2009), independent component analysis (Bach and Jordan, 2002), probabilistic graphical models (Koller and Friedman, 2009) and causal inference (Pearl, 2000; Spirtes et al., 2000).

In principle, testing for conditional independence is quite straightforward: Obtain a sample S of i.i.d. observations drawn from a distribution P_{xyz} , and split it evenly into two subsamples S_1 and S_2 of equal size. Leaving S_1 intact, permute S_2 so as to simulate a sample from $P_{xz}P_{y|z}$ (i.e., $X \perp\!\!\!\perp Y \mid Z$); apply a two-sample test to determine if S_1 and S_2 are different. If the null hypothesis $P_{xyz} = P_{xz}P_{y|z}$ cannot be rejected, we fail to reject conditional independence; this does not establish it. Repeated random splits can reduce Monte Carlo variation, but calibration of an aggregated test requires separate justification.

In order for the preceding approach to CI testing to be feasible, we need an effective two-sample test to determine if S_1 and S_2 are different. Gretton et al. (2012) introduced a framework for designing such tests using a well-behaved (e.g., smooth) function, which is large on the points drawn from one distribution and small on the points from the other distribution. The framework uses as test statistic, MMD, the largest difference between the mean function values on the two samples; when MMD is large, the samples are likely from different distributions. In order for MMD to be effective in practice, the class of functions used to define it should be (i) rich enough to ensure that the MMD vanishes if and only if the two distributions being compared are identical; and (ii) sufficiently restricted so as to ensure that the empirical estimate of MMD converges quickly to its expected value as the sample size is increased. As shown by Gretton et al. (2012), a bounded characteristic kernel provides an RKHS unit ball that distinguishes probability distributions and permits concentration bounds for empirical

1 INTRODUCTION

Random variables X and Y are said to be conditionally independent given Z , denoted by $X \perp\!\!\!\perp Y \mid Z$, if a joint probability distribution $P_{xyz} = P_{xz}P_{y|xz}$

MMD under the stated sampling assumptions.

Doran et al. (2014) introduced the Kernel Conditional Independence Permutation Test (KCIPT), using MMD in a kernel-induced feature space as the test statistic for determining whether $X \perp\!\!\!\perp Y \mid Z$. Doran et al. (2014) evaluated several kernel-based independence tests based on how well a test correctly rejects the null hypothesis $P_{xyz} = P_{xz}P_{y|z}$ by estimating the *power* of the test (as measured by the Area Under the Power Curve) and the *calibratedness* of the test, i.e., the extent to which it accurately estimates the probability distribution of the test statistic under the null hypothesis (as measured by a Kolmogorov-Smirnov divergence of the observed distribution of p-values from the uniform distribution). Using the preceding procedure, with the number of bootstraps B set equal to 25, Doran et al. (2014) asserted that KCIPT “has power competitive with existing kernel-based approaches” and that it is well-calibrated compared to other kernel independence tests such as KCIT (Zhang et al., 2011) and CHSIC (Fukumizu et al., 2008).

Increasing the number of bootstraps can change both rejection rates and calibration. We therefore examine how these quantities vary with B and find that calibration degrades in the examples considered. This suggests that the observed increase in rejection rates of KCIPT comes at the expense of its calibratedness. Based on our analysis of the limitations of KCIPT, we propose a new CI test, which we call the Self-Discrepancy CI Test (SDCIT). SDCIT modifies the usual unbiased squared-MMD estimator and approximates a null distribution using half-sampling without replacement. We present results of experiments that demonstrate several advantages of SDCIT over the existing kernel-based CI tests.

2 PRELIMINARIES

We mostly follow the notational conventions of Doran et al. (2014): We use upper case letters, e.g., X , to denote random variables, and the corresponding lowercase letters $x \in \mathcal{X}$ to denote an instantiation of X in its domain $\mathcal{X}$. We use bold lowercase letters to denote sets (or vectors) of instantiations, e.g., $\mathbf{x} = (x_i)_{i=1}^n$. We denote by $x \sim \mathbf{x}$ the fact that x is an observation of a random variable sampled from the empirical distribution constructed from the finite sample $\mathbf{x}$. We denote by $\binom{\mathbf{x}}{\ell}$ a set of all ℓ -sized subsets of $\mathbf{x}$. Let $\pi(n)$ be the set of all permutations of $(1, 2, \dots, n)$ with an additional constraint that every permutation $\pi \in \pi(n)$ satisfies that $\forall_{i=1}^n \pi(i) \neq i$. We denote the application of a permutation π to a sequence $\mathbf{y}$ by $\pi\mathbf{y} := (y_{\pi(i)})_{i=1}^n$. We use μ , the average function, to denote the average of a given parameter, e.g., a vector, set, or matrix.

2.1 KERNEL TWO-SAMPLE TEST

Let k_x be a kernel function $k_x : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space induced by a kernel k . For instance, $k_x(x', x'') = \langle \phi_x(x'), \phi_x(x'') \rangle_{\mathcal{H}_{k_x}}$ where ϕ_x is a feature mapping from $\mathcal{X}$ to $\mathcal{H}_{k_x}$. We use $\mathbf{K}$ to denote a Gram matrix (i.e., kernel matrix) corresponding to a kernel function k . Thus, $\mathbf{K}_x(\mathbf{x}, \mathbf{x}')_{ij} := k_x(x_i, x'_j)$. We use k_{xy} to denote the product of kernels k_x and k_y and define it as $k_{xy}((x_i, y_i), (x_j, y_j)) := k_x(x_i, x_j)k_y(y_i, y_j)$. Kernel mean embedding (KME) allows us to represent a probability distribution as a point in the kernel-induced Hilbert space (Berlinet and Thomas-Agnan, 2004; Smola et al., 2007; Sriperumbudur et al., 2010), $\mu_P := \int k(\cdot, x)P(dx) \in \mathcal{H}_k$. Maximum mean discrepancy (MMD) is an integral probability metric that provides a measure of distance between the two probability distributions P and Q : $\text{MMD}^2[\mathcal{F}, P, Q] := \left[\sup_{f \in \mathcal{F}} (\mathbb{E}_{x \sim P} f(x) - \mathbb{E}_{x \sim Q} f(x)) \right]^2 = \|\mu_P - \mu_Q\|_{\mathcal{H}_{k_x}}^2$ where $\mathcal{F}$ is a unit ball in $\mathcal{H}_{k_x}$. Given a characteristic kernel, $\text{MMD}(P, Q)$ is 0 if and only if distributions $P = Q$ (Sriperumbudur et al., 2010). Given two independent samples of the same size, $\mathbf{x}^{(1)} = \{x_i^{(1)}\}_{i=1}^m$ and $\mathbf{x}^{(2)} = \{x_i^{(2)}\}_{i=1}^m$, let $u_i := (x_i^{(1)}, x_i^{(2)})$. Then, an empirical unbiased estimate of squared MMD

$$\text{MMD}_u^2(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) := \frac{1}{m(m-1)} \sum_{i \neq j=1}^m h(u_i, u_j) \quad (1)$$

is a U-statistic in the paired observations u_i where

$$h(u_i, u_j) := k(x_i^{(1)}, x_j^{(1)}) + k(x_i^{(2)}, x_j^{(2)}) - k(x_i^{(1)}, x_j^{(2)}) - k(x_j^{(1)}, x_i^{(2)}).$$

A kernel two-sample test (Gretton et al., 2007, 2009, 2012) uses MMD as the test statistic and its distribution under the null hypothesis to test for homogeneity. The distribution of MMD can be estimated in a number of ways including moment-based approximation (Gretton et al., 2012), Gram matrix spectrum (Gretton et al., 2009), and resampling procedures. For instance, KCIPT uses a bootstrap procedure to repeatedly measure the MMD between two equal-sized random subsamples of the union of the two given samples.

3 KCIPT

We now proceed to describe KCIPT and discuss its limitations that motivate our proposal for SDCIT. Suppose we are given a sample $\Omega = \{(x_i, y_i, z_i)\}_{i=1}^n = (\mathbf{x}, \mathbf{y}, \mathbf{z})$ of n observations drawn i.i.d. from P_{xyz} . We split Ω randomly into two subsamples of equal size: $\Omega^{(1)} =$

Algorithm 1 KCIPT (Doran et al., 2014) with an *unbiased* estimator of squared MMD as its starting point.

Input: B : the number of bootstraps, b : the size of sample for a null distribution per bootstrap, M : the size of sample for a bootstrap null distribution, Ω : a sample $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ of size n

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1: for  $i \in 1 \dots B$  do
2:    $\Omega_i^{(1)}, \Omega_i^{(2)} \leftarrow$  randomly split  $\Omega$  evenly
3:    $\pi \leftarrow$  learn a permutation given  $\delta$  and  $\mathbf{z}_i^{(2)}$ 
4:    $\Omega_{i,\pi}^{(2)} \leftarrow$  apply  $\pi$  to  $Y$  values of  $\Omega_i^{(2)}$ 
5:    $T_i \leftarrow \text{MMD}_u^2(\Omega_i^{(1)}, \Omega_{i,\pi}^{(2)})$ 
6:   for  $j \in 1 \dots b$  do
7:      $\Omega', \Omega'' \leftarrow$  randomly split  $\Omega_i^{(1)} \cup \Omega_{i,\pi}^{(2)}$  evenly
8:      $(\mathbf{N})_{ij} \leftarrow \text{MMD}_u^2(\Omega', \Omega'')$ 
9:  $T \leftarrow$  average of  $\{T_i\}_{i=1}^B$ 
10:  $\mathbf{N} \leftarrow \{\text{average of } \{s_{ij}\}_{i=1}^B \mid \forall_{i=1}^B s_{ij} \sim \mathbf{N}_i\}_{j=1}^M$ 
11: return a p-value of  $T$  from  $\mathbf{N}$ 

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$(\mathbf{x}^{(1)}, \mathbf{y}^{(1)}, \mathbf{z}^{(1)})$ and $\Omega^{(2)} = (\mathbf{x}^{(2)}, \mathbf{y}^{(2)}, \mathbf{z}^{(2)})$. The second split sample $\Omega^{(2)}$ is permuted so as to simulate a sample from $P_{xz}P_{y|z}$.¹ Such permutation $\pi \in \pi(n/2)$ must satisfy that $z_i^{(2)} = z_{\pi(i)}^{(2)}$ to preserve the relationship between Y and Z while breaking ties between X and Y . In many interesting cases, including $\mathcal{Z} = \mathbb{R}$, $\mathbf{z}$ often consists of unique values and such a permutation may not exist. Hence, under the assumption $P(Y | z) \approx P(Y | z')$ if $z \approx z'$, we relax the requirement to learn a permutation $\pi \in \pi(n/2)$ by minimizing $\delta(\mathbf{z}^{(2)}, \pi \mathbf{z}^{(2)})$ where δ is a user-defined distortion measure. A common choice for δ is the sum of distances between permuted $\mathbf{z}$ values, $\sum_{i=1}^{n/2} d_z(z_i^{(2)}, z_{\pi(i)}^{(2)})$ where d_z is induced by k_z , i.e., $d_z^2(z', z'') := k_z(z', z') + k_z(z'', z'') - 2k_z(z', z'')$ or a regression-based distance $d_z(z', z'') := \|f(z') - f(z'')\|_2$ where f is a function relating Z and Y that can be learned, for instance, using a Gaussian process regression (GPR) (Zhang et al., 2011; Doran et al., 2014).

We apply the learned permutation π to $\mathbf{y}^{(2)}$ in $\Omega^{(2)}$ to obtain $\Omega_\pi^{(2)} := (\mathbf{x}^{(2)}, \pi \mathbf{y}^{(2)}, \mathbf{z}^{(2)})$. We then perform a two-sample test between $\Omega^{(1)}$ and $\Omega_\pi^{(2)}$ where we compute a p-value using $\text{MMD}_u^2(\Omega^{(1)}, \Omega_\pi^{(2)})$ as the test statistic. We approximate the null distribution of the test statistic by repeatedly measuring MMD_u^2 between fresh random splits of the pooled samples (see lines 6–8).

We can repeat the random splitting and average the resulting statistics. Suppose we bootstrap the aforementioned two-sample test B times. Let T_i and $\mathbf{N}_i$ be the i th MMD_u^2 estimate and a corresponding null distribution (called the ‘inner’ null distribution). The bootstrapped test statistic is simply the average of test statis-

¹One can either permute X or Y . For consistency, we will stick with permuting Y as in Doran et al. (2014).

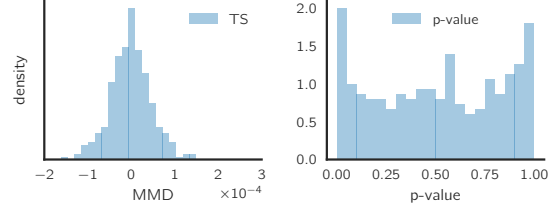


Figure 1: The distribution of test statistics of KCIPT with $B = 5000$ on 300 samples generated under the null hypothesis and the corresponding p-value distribution. Section 5.4 discusses an illustrative, data-dependent reference-variance match at $B = 2145$. The generating Hénon lag coefficient is unverified; see Section 5.1.

tics over each of the splits, i.e., $\mu(\{T_i\}_{i=1}^B)$. The null distribution of the bootstrapped test statistic is obtained using a Monte Carlo simulation by averaging together the draws from each individual statistic’s null distribution, $\{\mu(\{s_{ij}\}_{i=1}^B) \mid \forall_{i=1}^B s_{ij} \sim \mathbf{N}_i\}_{j=1}^M$, where M is the number of points drawn from each of the B null distributions.

Doran et al. (2014) evaluated three kernel-based CI tests, KCIT, CHSIC, and KCIPT, based on how well a test correctly rejects the null hypothesis (power) and how similar the distribution of p-values under the null hypothesis is to the uniform distribution between 0 and 1 (calibratedness). Doran et al. (2014) concluded that KCIPT “has power competitive with existing kernel-based approaches” and that it is well-calibrated.

3.1 LIMITATIONS OF KCIPT

Larger B reduces variation over random splits of a fixed dataset, but it does not guarantee monotonically increasing power. To see the calibration concern, condition on a dataset Ω and write $\theta(\Omega) = \mathbb{E}[T_i \mid \Omega]$. The average statistic converges to $\theta(\Omega)$ as B increases. Independently averaging draws from the inner null distributions instead gives a reference variance of order B^{-1} , with a limiting center near zero for the squared-MMD U-statistic. The variation of $\theta(\Omega)$ across newly sampled datasets need not vanish. If $\theta(\Omega) > 0$, the shrinking reference can yield rejection even for a dataset drawn under the null. The covariance between split statistics induced by reusing Ω is not reproduced by independent draws from the inner reference distributions. This concerns increasing B at fixed sample size, not an asymptotic claim about increasing n . See Figure 1, where KCIPT with $B = 5000$ results in a type I error rate of around 10% given $\alpha = 0.05$. We will examine this phenomenon more closely later in Section 5.

Different random seeds can yield different splits and hence different p-values for a fixed dataset. Their condi-

Algorithm 2 MMSD

Input K_{xz}, K_y : Gram matrices for $\{(x_i, z_i)\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$;
D: a pairwise distance matrix for $\mathbf{z}$; $n > 3$

- 1: $\pi \leftarrow \text{find } \pi \in \pi(n) \text{ minimizing } \delta \text{ given } D$
 - 2: $\mathcal{J} \leftarrow \{(i, j) \mid 1 \leq i \neq j \leq n, i \neq \pi(j), j \neq \pi(i)\}$
 - 3: $K_1, K_2, K_3 \leftarrow K_{xz} \odot K_y, K_{xz} \odot K_y[\pi, \pi], K_{xz} \odot K_y[:, \pi]$
 - 4: $T \leftarrow |\mathcal{J}|^{-1} \cdot \sum_{(i,j) \in \mathcal{J}} (K_1 + K_2 - K_3 - K_3^\top)_{i,j}$
 - 5: **return** $T, \pi, \mathcal{J}$
-

tional distribution need not be uniform, and this variability alone does not imply miscalibration. Calibration concerns repeated datasets generated under the null, whereas a fixed seed addresses computational reproducibility. The experiments in Section 5.4 illustrate both phenomena. Selecting B thus requires attention to Monte Carlo variability and calibration; neither a small nor a large B is universally appropriate.

4 SELF-DISCREPANCY CI TEST

We proceed to introduce Self-Discrepancy Conditional Independence Test (SDCIT), a new permutation and kernel-based CI test, using a new test statistic MMSD, which is based on an unbiased squared MMD estimate. We specify the empirical distribution of the MMSD and that under the null hypothesis. Under the explicit sufficient conditions in Theorem 1, MMSD converges to a population discrepancy that vanishes if and only if conditional independence holds. This does not establish the calibration or power consistency of the resulting p-value procedure.

4.1 MAXIMUM MEAN SELF-DISCREPANCY

MMSD uses the full sample Ω rather than random splits of it. For a fixed permutation it is deterministic. The learned permutation, tie-breaking, and the resampled p-value can still depend on a random seed. We replace MMD_u^2 estimated from the samples obtained by randomly splitting Ω into two parts where one part is left intact and the other is permuted to break the ties that violate conditional independence by a variant of MMD_u^2 between the given sample Ω and its permuted counterpart. We learn a permutation π of Ω by minimizing distortion measure δ on $\mathbf{z}$. Let Ω_π be a sample where the permutation π is applied to $\mathbf{y}$ of Ω . However, because Ω_π is *not* obtained independently from Ω (see Equation 1), we cannot naively measure MMD_u^2 between Ω and Ω_π . Hence, we introduce a new test statistic, which we call Maximum Mean Self-Discrepancy (MMSD), which estimates the discrepancy between a sample Ω and its conditionally independent counterpart Ω_π by omitting terms whose two kernel arguments reuse an original observa-

tion. This does not remove all dependence induced by learning the permutation, or make the resulting statistic exactly unbiased. Let

$$h(i, j) := k_{xyz}((\Omega)_i, (\Omega)_j) + k_{xyz}((\Omega_\pi)_i, (\Omega_\pi)_j) - k_{xyz}((\Omega)_i, (\Omega_\pi)_j) - k_{xyz}((\Omega_\pi)_j, (\Omega)_i)$$

where $(\cdot)_i$ represents the i th observation of the argument. We omit $i = j$, $i = \pi(j)$, and $j = \pi(i)$, because these indices make the kernel arguments share an original observation. Thus, for a derangement π , let

$$\mathcal{J} = \{(i, j) : 1 \leq i \neq j \leq n, i \neq \pi(j), j \neq \pi(i)\}.$$

This is an index-exclusion rule. Conditioning on a learned permutation alone does not imply that the retained observations are independent. Based on the extent to which the permutation is reciprocal (i.e., $i = (\pi \circ \pi)(i)$), the size of $\mathcal{J}$ ranges from $n^2 - 3n$ to $n^2 - 2n$. We proceed to estimate MMSD as follows:

$$T := \frac{1}{|\mathcal{J}|} \sum_{(i,j) \in \mathcal{J}} h(i, j). \quad (2)$$

MMSD uses a single learned permutation for its raw statistic. The KCIPT convergence argument (Doran et al., 2014, Supplementary Appendix A), together with an $O(n^{-1})$ term-removal bound, gives the following result.

Theorem 1. *Let $(X_i, Y_i, Z_i)_{i=1}^n$ be i.i.d. on compact metric spaces. Fix bounded continuous real-valued positive-definite kernels k_x, k_y, k_z whose product k_{xyz} is universal, and let their feature norms be bounded by C_x, C_y, C_z . Set $C = C_x C_y C_z$ and $d_z(z, z') = \|\phi_z(z) - \phi_z(z')\|$. Let π_n be a derangement selected using only $(Z_1, \dots, Z_n)$ and auxiliary randomness independent of the observations. If*

$$\delta_n = \frac{1}{n} \sum_{i=1}^n d_z(Z_i, Z_{\pi_n(i)}) \xrightarrow{P} 0,$$

then, for $n > 3$, the MMSD statistic in Equation 2 satisfies

$$T_n \xrightarrow{P} \|\mu_P - \mu_Q\|^2,$$

where

$$Q(dx, dy, dz) = P_{XZ}(dx, dz)P_{Y|Z}(dy | z).$$

The limit is zero if and only if $X \perp\!\!\!\perp Y | Z$.

Proof. Appendix C proves that the permuted empirical embedding converges to μ_Q . The original embedding converges to μ_P . The squared distance between these embeddings differs from MMSD by $O(n^{-1})$, including the change in normalization when terms are omitted. Characteristicness then gives both directions of the claim. $\square$

Algorithm 3 EMPMMSD

Input K_{xz}, K_y : Gram matrices for $\{(x_i, z_i)\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$; D : a pairwise distance matrix for $\mathbf{z}$; b : a number of samples to be generated.

- 1: $\mathbf{T}' \leftarrow$ initialize a vector of size b .
 - 2: **for** $r = 1, \dots, b$ **do**
 - 3: $\mathbf{i} \leftarrow$ randomly choose $\frac{n}{2}$ unique integers from $(1, \dots, n)$
 - 4: $\mathbf{T}'_r, \cdot, \cdot \leftarrow \text{MMSD}(K_{xz}[\mathbf{i}, \mathbf{i}], K_y[\mathbf{i}, \mathbf{i}], D[\mathbf{i}, \mathbf{i}])$
 - 5: **return** $\frac{1}{2} \cdot (\mathbf{T}' - \mu(\mathbf{T}')) + \mu(\mathbf{T}')$
-

The Z -only selection describes the RKHS-distance permutation rule. No continuity assumption on the conditional feature means is needed. The result does not justify a same-sample regression distance, adaptive bandwidths, or a half-sampling null distribution without further analysis. For a product of unit-amplitude Gaussian kernels, one can take $C = 1$. We describe the pseudocode for MMSD estimate in Algorithm 2 where $\mathbf{M}[\mathbf{a}, \mathbf{b}]$ denotes a submatrix of $\mathbf{M}$ where its rows and columns are selected and ordered by $\mathbf{a}$ and $\mathbf{b}$, respectively, that is, $(\mathbf{M}[\mathbf{a}, \mathbf{b}])_{i,j} = M_{a_i, b_j}$. A colon ‘:’ represents entire rows or columns. An operator $\odot$ denotes a Hadamard product (i.e., element-wise multiplication).

4.2 EMPIRICAL DISTRIBUTION OF MMSD

We now turn our attention to obtaining the empirical distribution of MMSD, the test statistic used by SDCIT. Although the distribution of MMD_u^2 has been thoroughly analyzed (Gretton et al., 2012), it is not at all immediately obvious how the analysis of MMD_u^2 , which is based on two independent samples, is applicable to the case of MMSD, which is defined with respect to two *dependent* samples. Unlike other point estimates such as mean, MMSD on a bootstrap sample can be problematic: it is possible that pairs of the repeated observations in a bootstrap sample are permuted to each other thereby violating the conditional independence desired in the permuted sample. The resulting test statistic from such bootstrap sample will be closer to 0 than it should be.

We instead sample without replacement. Results on bagged estimators (Buja and Stuetzle, 2006; Friedman and Hall, 2007) motivate half-sampling, but do not establish the sampling distribution of this learned-permutation statistic. Let $\Omega' = \{\Omega'_r\}_{r=1}^b$ consist of independent draws of subsets of size $n/2$ from the observed sample, and compute $T'_r = \text{MMSD}(\Omega'_r, D[\Omega'_r, \Omega'_r])$. Algorithm 3 uses the rescaling

$$\mathbf{T} = \frac{1}{2}(\mathbf{T}' - \mu(\mathbf{T}')) + \mu(\mathbf{T}').$$

The factor $1/2$ is retained to describe the original procedure. Counting pairwise terms alone does not derive this factor: their dependence and the conditional resam-

pling distribution must be taken into account. The resulting procedure is supported empirically by the calibration experiments in Section 5; a general theoretical justification of the null approximation is not supplied.

4.3 APPROXIMATE NULL DISTRIBUTION OF MMSD

An approximate null distribution of test statistic can often be obtained by applying the statistic on many samples generated under the null hypothesis ($P_{xz}P_{y|z}$ in this case). Since we have access to neither $P_{xz}P_{y|z}$ nor the model for the distribution, we will apply half-sampling without replacement on the permuted sample Ω_π to approximate the null distribution of MMSD. As described in the previous section, we generate Ω'_π , b half-samples of Ω_π via sampling without replacement. However, we need to be careful in obtaining the test statistic from the resulting half samples. Let a half-sample $\Omega'_\pi = \{(x'_c, y'_c, z'_c)\}_{c=1}^{n/2} \in \Omega'_\pi$, where $\Omega'_\pi \subset \Omega_\pi$, contain the following two observations at its a th and b th indices, $(x'_a, y'_a, z'_a) = (x_i, y_{\pi(i)}, z_i) = (x_i, y_j, z_i)$ and $(x'_b, y'_b, z'_b) = (x_k, y_{\pi(k)}, z_k) = (x_k, y_i, z_k)$ where indices i, j , and k are originally from Ω . We can infer that $z_j \approx z_i \approx z_k$. Hence, a permutation π' to be learned for Ω'_π will likely permute y ’s in the two observations. If that is the case (i.e., $\pi'(a) = b$), then the permuted half-sample $(\mathbf{x}', \pi' \mathbf{y}', \mathbf{z}')$ will contain the observation $(x'_a, y'_{\pi'(a)}, z'_a) = (x_i, y'_b, z_i) = (x_i, y_i, z_i)$. That is, applying the permutation π' recovers an original observation in Ω failing to simulate conditional independence in $\Omega'_{\pi\pi'}$. Hence, the following constraints should be imposed in computing a permutation for Ω'_π , $\pi(f(\pi'(c))) \neq f(c)$ for every $1 \leq c \leq n/2$ where f is a mapping between the index of an observation in a subsample to that in Ω_π . In the preceding example, $\pi'(a) = b$, $f(b) = k$, $\pi(k) = i$, and $f(a) = i$. In other words, simply disallow permutations that associate y_i back to $(x_i, \cdot, z_i)$. In addition to constraints on the allowed permutations, identifying pairs that reuse an original observation requires tracking the composition of every observation in the half-samples of Ω_π .

To discourage reconstruction, define the off-diagonal excluded pairs

$$\mathcal{J}_\pi = \{(i, j) : i \neq j, i = \pi(j) \text{ or } j = \pi(i)\}.$$

These pairs are outside $\mathcal{I}$. The original text incorrectly penalized pairs in $\mathcal{J}$, reversing the intended mask. Setting $D_{ij} = \infty$ on $\mathcal{J}_\pi$ would impose hard exclusions, but can leave a half-sample with no feasible derangement. The released implementation instead uses the finite penalty

$$D_{ij}^\pi = D_{ij} + \lambda \mathbf{1}\{(i, j) \in \mathcal{J}_\pi\}, \quad \lambda = \max_{a,b} D_{ab}.$$

Algorithm 4 SDCIT

Input K_{xz}, K_y : Gram matrices for $\{(x_i, z_i)\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$; D : a pairwise distance matrix for $\mathbf{z}$; b : the size of empirical null distribution.

- 1: $T, \pi, J \leftarrow \text{MMSD}(K_{xz}, K_y, D)$
 - 2: $D^\pi \leftarrow \text{penalize } J_\pi \text{ as in Section 4.3}$
 - 3: $\cdot, \pi_2, J_2 \leftarrow \text{MMSD}(K_{xz}, K_y, D^\pi)$
 - 4: $D^{\pi_2} \leftarrow \text{penalize } J_{\pi_2} \text{ in the original } D$
 - 5: $\mathbf{N} \leftarrow \text{EMPMMSD}(K_{xz}, K_y[\pi_2, \pi_2], D^{\pi_2}, b)$
 - 6: **return** $\hat{p}$ using the tie-aware rule in Section 4.4.
-

The diagonal is excluded separately by the derangement requirement. This penalty discourages reuse; it does not forbid it. In particular it vanishes when $D = 0$. Algorithm 4 describes this implemented heuristic. Centering its reference distribution at zero removes the empirical mean but does not in itself establish unbiasedness or calibration.

4.4 ALGORITHM

Algorithm 4 combines MMSD with its approximate reference distribution. It first computes the observed statistic, then learns a second permutation π_2 with the finite penalty for π , and generates half-samples using the penalty for π_2 . This extra permutation is a heuristic intended to reduce the discrepancy between full-sample and half-sample matching quality. No ordering of their errors is guaranteed. For an even $n \geq 8$, every half-sample has more than three observations, so its MMSD denominator is nonzero. The implementation may impose further requirements on its matching routine. The input distance matrix must be finite; the pseudocode does not require passing infinities to that routine. For clarity, this corrected version uses the tie-aware upper-tail estimate²

$$\hat{p} = \frac{1 + \sum_{r=1}^b \mathbf{1}\{N_r - \mu(\mathbf{N}) \geq T\}}{b + 1}.$$

The historical implementation instead uses $b^{-1} \sum_r \mathbf{1}\{N_r - \mu(\mathbf{N}) > T\}$. That strict-tail rule returns zero even when the statistic and all reference values equal zero, as occurs for constant data. The revised convention avoids this defect and zero Monte Carlo estimates; it does not make a heuristic reference distribution an exact or asymptotically valid null distribution. The retained numerical results used the historical implementation and have not been recomputed with this convention. An additional empirical adjustment appears in Appendix B; the basic pseudocode omits it. Table 2 compares the procedures.

²Code snapshot for the proposed p-value and native-code corrections: revision a4c990d. The exact revision used for the historical benchmarks has not been established.

5 EXPERIMENTS

Experimental scope of this correction. Experiments were not rerun; reported experimental numerical values and figure assets are unchanged. The revised p-value rule and corrections to native matrix shuffling and matching-cost arithmetic can change computed outputs. Their effects on the historical AUPC, KS, type I error, and run-time comparisons have not been established. The results below report the original experiments, not a validation of the corrected implementation.

We compare the performance of SDCIT with that of other kernel-based CI tests, including KCIT, CHSIC, and KCIPT, with respect to the following two criteria: (i) Area Under Power Curve (AUPC) which estimates how powerful the test is by measuring the area under the cumulative distribution function (CDF) of $\mathbf{p}$, a set of p-values produced by running a test on a set of samples; and (ii) Kolmogorov-Smirnov (KS) statistic (a measure of the largest discrepancy between cumulative distribution functions of the two distributions), applied to $\mathbf{p}$ and a uniform distribution to assess the degree to which the p-values produced by the test given samples under the null hypothesis are distributed uniformly in the interval $[0, 1]$. We also compare the different tests with respect to their type I error rates (given a significance level $\alpha = 0.05$) and their run time. We examine the robustness of the different tests with respect to the choice of the kernel function. We contrast SDCIT with KCIPT when the latter uses a large number of bootstraps.

We implemented SDCIT and KCIPT. The archived 2017 experiment scripts call SDCIT with its default Appendix B adjustment enabled; a separate script produces the unadjusted comparison. The retained figures have not been regenerated, and an exact run-to-figure provenance record was not available for every asset.³ Unlike the original implementation of KCIPT which employs a simplex algorithm to learn permutations, all permutations are *approximately* learned using minimum cost perfect matching algorithm (BLOSSOM-V, Kolmogorov, 2009) with heuristics for local improvement where three 2-cycles are transformed to two 3-cycles and a 2-cycle and a 3-cycle are transformed to a 5-cycle. Gaussian RBF kernel $k(v, w) := \exp(-\|v - w\|_2^2 / (2\sigma^2))$ is used across all experiments where σ is determined by the median heuristic (Gretton et al., 2005). SDCIT uses empirical null distributions of size 10^3 . We used the recommended settings for KCIPT (i.e., $B = 25$ and $b, M = 10^4$) and for CHSIC. For KCIT, we used *both* the original implementation, in which variables are normalized to unit variance and kernel parameters are set empirically (Zhang et al., 2011),

³The code is available online at <https://github.com/sanghack81/SDCIT>

and a modified implementation, where the variables are not normalized and the kernel parameters are set by median heuristic. For both implementations, GPR is used to optimize the regularization parameter.

5.1 EXPERIMENTAL SETTING

Following previous work (Fukumizu et al., 2008; Zhang et al., 2011; Doran et al., 2014), we conducted experiments on two synthetic datasets, post-nonlinear noise and chaotic time series based on the models used in Doran et al. (2014), where we study two types of setting, i.e., $X \perp\!\!\!\perp Y \mid Z$ and $X \not\perp\!\!\!\perp Y \mid Z$.

Post-nonlinear noise data is generated using the model described by Zhang and Hyvärinen (2009); Zhang et al. (2011). X and Y are constructed from functions of the form $G_X(F_X(Z_1) + E_X)$ and $G_Y(F_Y(Z_1) + E_Y)$, respectively, where G and F are smooth nonlinear functions, E_X and E_Y are independent Gaussian errors independent of Z , and Z_1 is a component of an m -dimensional conditioning variable $Z := \{Z_i\}_{i=1}^m$. The conditioning variable Z satisfies that $(Z_2, \dots, Z_m) \perp\!\!\!\perp (X, Y, Z_1)$ for some m making only Z_1 relevant to X and Y . Since $X \perp\!\!\!\perp Y \mid Z_1$ by construction, we also generate samples under the alternative, $X \not\perp\!\!\!\perp Y \mid Z$, by adding identical Gaussian noise to both X and Y .

Chaotic time series is based on the coupled Hénon map. Each X_t and Y_t is four-dimensional, $X_t = (X_{i,t})_{i=1}^4$ and $Y_t = (Y_{i,t})_{i=1}^4$ where

$$\begin{aligned} X_{1,t} &:= 1.4 - X_{1,t-1}^2 + 0.3X_{2,t-1} \\ Y_{1,t} &:= 1.4 - \gamma X_{1,t-1}Y_{1,t-1} - (1 - \gamma)Y_{1,t-1}^2 \\ &\quad + b_Y Y_{2,t-1} \end{aligned}$$

where $X_{2,t}$ and $Y_{2,t}$ inherit $X_{1,t-1}$ and $Y_{1,t-1}$, respectively. The published equations in Doran et al. (2014) and the original version of this paper specify $b_Y = 0.3$, as does the SDCIT Python generator. However, the public KCIPT MATLAB generator and the archived SDCIT MATLAB generator use $b_Y = 0.1$. The archived experiment scripts load MATLAB-generated data. We record this discrepancy rather than assert an exact match between the displayed model and every retained figure; the original generated datasets have not been checked in this correction. The third and fourth dimensions of X_t and Y_t correspond to Gaussian noise $\mathcal{N}(0, 0.5^2)$ to make the data more challenging. The parameter γ controls dependence. Regardless of γ , $X_{t+1} \perp\!\!\!\perp Y_t \mid X_t$ holds. For $\gamma > 0$, the benchmark uses the reverse direction as the dependent setting. Successive rows from a trajectory are not i.i.d.; these experiments are an empirical stress test outside the sampling assumptions of Theorem 1.

Following Doran et al. (2014), sets of 300 samples were

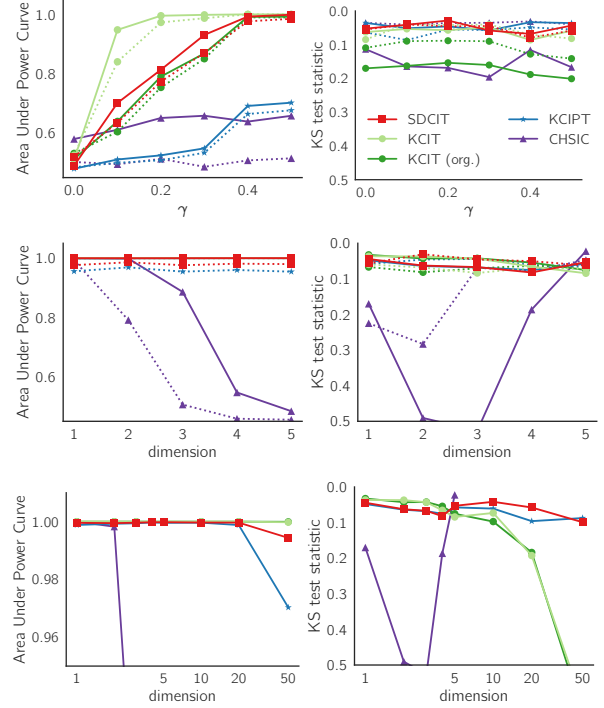


Figure 2: AUCP and KS statistics for **(Top)** Chaotic time series **(Middle)** Post-nonlinear noise data **(Bottom)** High-dimensional conditioning variables on post-nonlinear noise dataset. Dotted and solid lines are for the samples of size 200 and 400, respectively. In the bottom row, CHSIC results are included for dimensions 1–5 only; its AUCP values at dimensions 3–5 fall below the displayed range. The generating Hénon lag coefficient is unverfied; see Section 5.1.

generated for post-nonlinear noise data per each condition based on the combination of: two variants ($X \perp\!\!\!\perp Y \mid Z$ and $X \not\perp\!\!\!\perp Y \mid Z$), different dimensions $m \in \{1, 2, 3, 4, 5, 10, 20, 50\}$, and different sizes (200 and 400). Similarly, we generated sets of 300 samples of chaotic time series data for the two directions, two sample sizes, and γ ranging from 0 to 0.5 by 0.1. At $\gamma = 0$, both directions are null baselines; the reverse direction becomes the dependent setting when $\gamma > 0$. Datasets under the null hypothesis are used to report KS statistic and datasets under the alternative hypothesis are used to report AUCPs. SDCIT and KCIPT used RKHS distance for chaotic time series data and, for post-nonlinear noise data, regression-based distance is used where functions are learned based on GPR with automatic relevance determination.

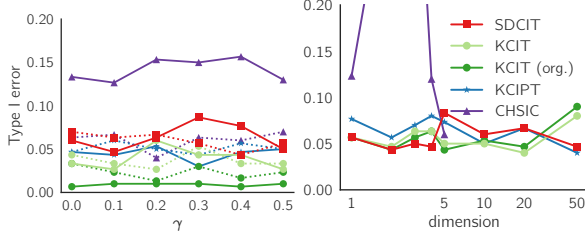


Figure 3: Type I error rates with $\alpha = 0.05$ (Left) chaotic time series data (Right) post-nonlinear noise data. The generating Hénon lag coefficient is unverified; see Section 5.1.

5.2 COMPARISONS AMONG KERNEL-BASED CI TESTS

Figure 2 illustrates the performance of the four tests on the two datasets under various conditions. Note that the original implementation of KCIT with recommended setting is labeled as KCIT (org.) in the figure. Plots in the left column represent AUPCs and those in the right column show KS statistic which measures the degree to which the distribution of p-values under the null hypothesis deviates from the uniform distribution. Note that the plots for the different tests may differ in the range of values plotted on the y-axis and the values on the y-axis for plots of KS statistic are in descending order.

With respect to power, SDCIT is ranked right below KCIT for both datasets although SDCIT is more powerful than the implementation of KCIT used in the experiments reported by Doran et al. (2014) for chaotic time series. On post-nonlinear noise data, all tests except CHSIC showed comparable power although the permutation-based methods (SDCIT and KCIPT) show a slight loss in power when $n = 200$. When the number of conditioning variables increases ($m = 50$), SDCIT slightly loses its power (AUPC=0.9944). In the case of KCIPT, we observed a far smaller loss in power (AUPC=0.9703) compared to that reported (around 0.79, Doran et al., 2014). In these experiments, SDCIT has AUPC comparable to or greater than the alternatives except KCIT. These comparisons must be read together with the null rejection rates. The hypotheses $P_{XYZ} = P_{X|Z}P_{Y|Z}P_Z$ and $P_{XY|Z} = P_{X|Z}P_{Y|Z}P_Z$ -almost surely are equivalent. They cannot explain differences between the tests. Different statistics, regularization, and null approximations may affect their performance.

Next, we compare the tests based on their calibratedness. For both datasets, SDCIT and KCIPT have very consistent null distributions with the KS statistics generally below 0.1. Also the results vary little with sample size. KCIT demonstrates difficulty obtaining accurate null distribu-

n	SDCIT $b = 500$	SDCIT $b = 1000$	KCIT
200	0.50 ± 0.02	0.96 ± 0.02	1.17 ± 0.19
400	2.41 ± 0.07	4.60 ± 0.07	4.68 ± 0.75

Table 1: Running time in seconds averaged over 300 samples of chaotic time series data (including time for kernel and distance matrices computation)

tions for CI tests with high-dimensional conditional variables. We also report the type I error rates on the two datasets in Figure 3. All tests except CHSIC and the KCIT (org.) reject around 5% of samples under the null hypothesis given a significance level α set to 0.05.

We report the run time of SDCIT and KCIT in Table 1. Based on a set of 300 samples of chaotic time series data, we measured the running time on an iMac with Core i7 3.5Ghz CPU allowing only a single thread. Learning permutations is a substantial computational step, but the $O(nm \log n)$ bound in the original text was misattributed to Blossom V. Kolmogorov (2009) discusses that bound for a different matching implementation. Table 1 reports measured times and does not establish a worst-case bound. In these measurements, SDCIT took less than five times as long when the sample size doubled. The run time is proportional to b , the size of empirical null distribution. Since SDCIT is trivially parallelizable, the run time can be significantly reduced by using multiple processor cores. Note that KCIPT is about 10 times slower even when $B = 25$ due to $B(b + 1)$ MMD computations.

5.3 ROBUSTNESS WITH RESPECT TO KERNEL CHOICE

The median heuristic to determine the kernel parameter for an RBF kernel works well for both SDCIT and KCIT in previous experiments. However, the kernel choice is entirely at the user’s discretion. A well-designed kernel-based test should be robust with respect to the choice of kernel parameters and should have a consistent null distribution. Consider a parametrized RBF kernel, $k^{(C)}(x, x') = \exp(-C\|x - x'\|^2/2\sigma^2)$, where σ^2 is chosen based on the median heuristic. We explore how KCIT and SDCIT behave as C varies from 10^{-2} to 10^2 . We report AUPC and KS statistics based on a set of 300 chaotic time series datasets with ‘dependent’ and ‘independent’ modes, respectively, both with $\gamma = 0.3$.

The original implementation of KCIT uses GPR to learn a regularization parameter only with a real vector Z . Hence, we also provide a result where an empirical kernel map for Z is approximately inferred from K_Z (la-

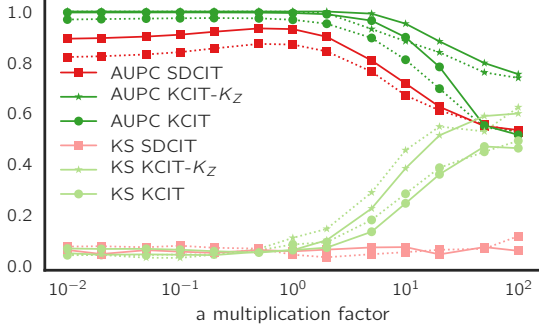


Figure 4: Changes of performance of KCIT and SDCIT with different choices of kernel parameters (relative to the default median heuristic). Solid and dotted lines represent $n = 400$ and 200 , respectively. The generating Hénon lag coefficient is unverified; see Section 5.1.

beled as KCIT- K_Z in Figure 4). For this benchmark, SDCIT is relatively stable across the tested Gaussian bandwidths, whereas KCIT loses calibration as C increases. This does not establish robustness to arbitrary kernel choices.

5.4 COMPARISON OF SDCIT AND KCIPT WITH INCREASING NUMBER OF BOOTSTRAPS

Changing B changes the Monte Carlo variability and reference distribution of KCIPT. Hence, we compare the performance of KCIPT with that of SDCIT as we increase the number of bootstraps used by KCIPT. For computational reasons, we replace the bootstrap procedure for the null distribution of KCIPT by an analytic one to cope with a large number of bootstraps (e.g., $B \geq 1000$). Since the inner null distributions are very similar to each other, we aggregated inner null distributions $\{\mathbf{N}_i\}_{i=1}^B$ where each distribution consists of a relatively small number of observations (100). We approximate the null distribution by $\mathcal{N}(0, \text{Var}(\bigcup_{i=1}^B \mathbf{N}_i)/B)$, as an approximation to line 10 of Algorithm 1. Conditional on the inner distributions, the variance of their independent average is exactly $B^{-2} \sum_i \text{Var}(\mathbf{N}_i)$; using the pooled variance divided by B additionally assumes their means and variances are sufficiently similar.

We used 300 samples of chaotic time series data of $n = 400$ and $\gamma = 0$. We selected $B = 2145$ by matching the reference variances for the first of these 300 datasets. This is an illustrative, data-dependent choice. It neither establishes equal power nor provides an independently validated tuning rule: all these datasets have $\gamma = 0$, so they do not measure power against conditional dependence. Equal variances also do not imply equal upper quantiles, especially for differently skewed distributions.

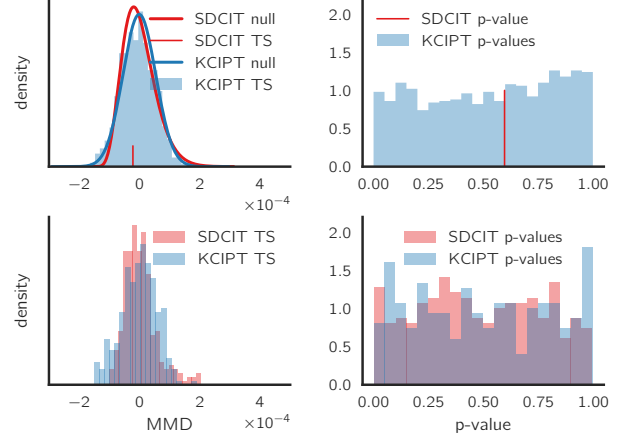


Figure 5: Empirical distributions and p-value distributions of SDCIT and KCIPT with $B = 2145$ based on the first sample (**Top**) and all 300 samples (**Bottom**) with $\gamma = 0.0$ and $n = 400$. The generating Hénon lag coefficient is unverified; see Section 5.1.

With the first sample, we compared empirical null distributions of SDCIT and KCIPT together with a test statistic for SDCIT and a distribution of test statistics for KCIPT through repeated trials with varying random seeds on the sample (recall the Monte Carlo variability of KCIPT) (see top of Figure 5). The null distribution of SDCIT is positively skewed (0.80), which is similar to the null distribution of MMD_u^2 in a two-sample test setting (Gretton et al., 2012) whereas KCIPT demonstrates normality because of large B . Since KCIPT test statistic is stochastic, the p-value is also stochastic. In this example, p-value is nearly uniformly distributed between 0 and 1. For other samples, we observed distributions of either negative or positive skew, shaped like staircases.

At the bottom of Figure 5, we plot test statistics of KCIPT and SDCIT on 300 samples. We observe that the p-values of SDCIT are fairly uniformly distributed, supporting calibration on this benchmark without establishing a general guarantee. However, KCIPT starts to lose its calibratedness as the number of bootstraps is increased, resulting in more p-values concentrated around the two extremes, 0 and 1. The analysis presented in Section 3.1 offers an explanation of the loss of calibratedness of KCIPT that is different from that conjectured by Doran et al. (2014), namely, an approximation error in representing $P_{xz}P_{y|z}$ due to imperfect permutations.

In Figure 6, with the 300 samples, we illustrate the null distribution of KCIPT and an empirical distribution of its statistics with $B = 20000$. For large B , Monte Carlo variation conditional on each dataset is small. Variation of the conditional mean across datasets can nevertheless remain, while the constructed reference variance de-

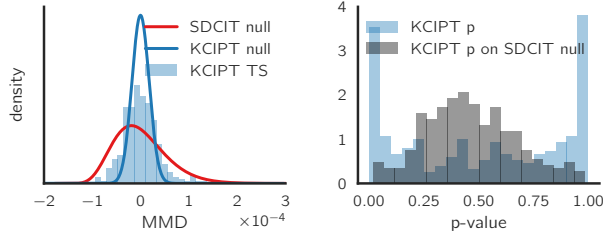


Figure 6: Null distributions of SDCIT and KCIPT, and an empirical distribution of test statistics of KCIPT with $B = 20000$ under the null hypothesis. The generating Hénon lag coefficient is unverified; see Section 5.1.

creases with B . This separation does not establish that a statistic accurately identifies conditional independence for each individual dataset. On the right side, we plot two p-value distributions of KCIPT against null distributions from KCIPT and SDCIT. The dome-shaped p-value distribution is similar to what we have observed, in another simulation, with SDCIT using a bagged statistic of MMSD (i.e., the average of empirical distribution of MMSD) as a test statistic without modifying the null distribution.

Our results suggest that SDCIT is not simply equivalent to KCIPT with some unknown but sufficiently large number of bootstraps B with respect to its power and calibratedness. The reported run-time comparison favors SDCIT for the settings tested. It does not establish a universal value of B at which the two tests have equal power or equal computational cost.

6 SUMMARY AND DISCUSSION

Doran et al. (2014) introduced KCIPT, which has the advantage of using one learned permutation per random split to reduce the conditional independence (CI) test to an easier two-sample test. Based on their experimental comparison of KCIPT with other kernel-based CI tests (CHSIC and KCIT), they concluded that KCIPT has power competitive with that of KCIT and is well-calibrated compared to other alternatives. We examined in greater detail the calibration loss with increasing bootstrap count that was also reported by Doran et al. (2014): KCIPT can lose calibration as the number of bootstraps increases. Careful analysis of the behavior of KCIPT leads us to propose SDCIT, a novel kernel-based CI test with a new test statistic based on modified MMD estimate, called Maximum Mean Self-Discrepancy. The raw statistic requires no outer bootstrap count for a fixed kernel, distance, and permutation rule. The p-value still depends on resampling and implementation choices. The results of our experiments with two benchmark datasets demonstrate

several advantages of SDCIT over other existing kernel-based CI tests. In the evaluated settings, SDCIT showed empirical calibration and competitive power. A general theoretical justification of its approximate null distribution remains open. Theorem 1 establishes consistency of the raw discrepancy statistic under explicit assumptions. It does not by itself establish finite-sample level control or consistency of the full test.

Acknowledgements

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A ALGORITHMIC COMPARISON

	SDCIT	KCIPT
H_0	$X \perp\!\!\!\perp Y \mid Z$	$X \perp\!\!\!\perp Y \mid Z$
T	modified MMD ²	averaged MMD ² statistics
null distribution	half-sampling without replacement	aggregated bootstrap null distributions
# of permutations	1 for T $b + 1$ for null.	B for T
# of MM(S)Ds	1 for T $b + 1$ for null.	B for T Bb for null.

Table 2: Comparison of SDCIT and KCIPT

B TAKING PERMUTATION ERROR INTO ACCOUNT

Our test statistic measures the distance between the original sample (representing P_{xyz}) and a pseudo-null sample (representing $P_{xz}P'_{y|z}$), where $P'_{y|z}$ approximates $P_{y|z}$. Ideally, the test statistic and its null distribution will be reliably estimated if permutation error is small and, hence, $P'_{y|z}$ approximates $P_{y|z}$ well.

We first relate an MMSD estimate and its corresponding permutation error during the estimate, and provide a means to adjust MMSD estimates. Let T be an MMSD estimate given K_{xz} , K_y , and D (see Algorithm 2). Let τ be an MMSD estimate assuming $K_x = \mathbf{1}_{n \times n}$, that is $\tau = \text{MMSD}(K_z, K_y, D)$ with the same permutation and exclusion mask used for T . Each half-sample pair (T_r, τ_r) also shares its sampled indices and permutation. The fit uses the rescaled outputs of Algorithm 3 for both coordinates; applying the same positive rescaling to both leaves the fitted slope unchanged. In other words, τ is the MMSD estimate between $P'_{y|z}P_z$ and $P_{y|z}P_z$. While T is, roughly, about the conditional dependence between X and Y given Z , τ measures permutation error, i.e., discrepancy between (y, z) and $(\pi y, z)$. We illustrate a null distribution $\{T_i\}_{i=1}^b$ and its associated $\{\tau_i\}_{i=1}^b$ in Figure 7. We can clearly observe that the distribution of τ is centered at 0 but still there are lots of null samples associated with non-negligible errors.

We then model T (under permutation error) as a function of unknown T^* (under zero permutation error) and τ . We assume a linear model $T = T^* + \beta\tau + \epsilon$ where ϵ is assumed to be zero-mean Gaussian noise. Given a null distribution $\{(T_i, \tau_i)\}_{i=1}^b$, we can fit a slope $\hat{\beta}$ by least squares with an intercept. If $\sum_i (\tau_i - \bar{\tau})^2 = 0$, we set $\hat{\beta} = 0$; this explicit convention agrees with the current Python implementation on constant error inputs. The released implementation uses $\beta = \max(0, \hat{\beta})$. Then, the null dis-

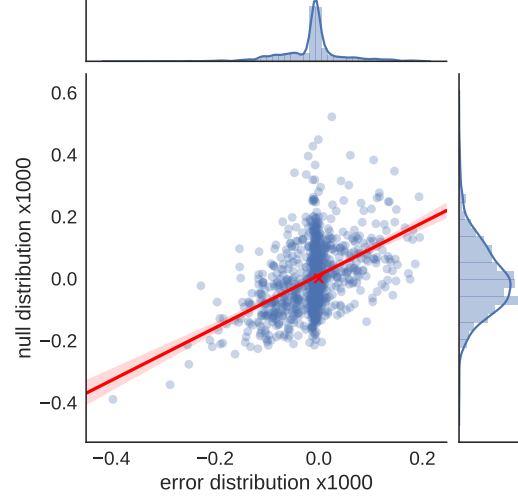


Figure 7: A null distribution and its corresponding errors measured with MMSD. A red cross near origin indicates the test statistic and its corresponding error. A red line indicates a fitted linear model.

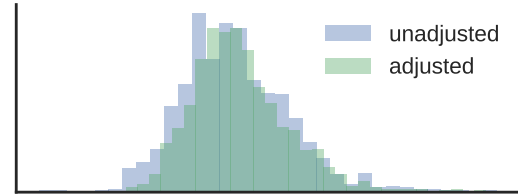


Figure 8: Unadjusted and adjusted null distributions.

tribution $\{T_i\}_{i=1}^b$ is adjusted to $\{T_i - \beta\tau_i\}_{i=1}^b$ and our test statistic is also adjusted similarly. The adjusted reference is centered before computing the p-value. Figure 8 illustrates a reduction in its empirical variance. This fitted adjustment is a heuristic; neither variance reduction nor a Gaussian working model proves improved power or calibration on new samples.

C PROOF OF THEOREM 1

Let $\mathcal{F}_n = \sigma(Z_1, \dots, Z_n, R_n)$, where R_n contains the independent auxiliary randomness used to select π_n . Conditioning on $\mathcal{F}_n$ fixes the permutation. Write $\pi = \pi_n$ and

$$W_i = \phi_x(X_i) \otimes \phi_y(Y_{\pi(i)}) \otimes \phi_z(Z_i).$$

Since $\pi(i) \neq i$, independence of distinct observations conditional on their Z values gives

$$\mathbf{E}[W_i \mid \mathcal{F}_n] = a_x(Z_i) \otimes m_y(Z_{\pi(i)}) \otimes \phi_z(Z_i),$$

where $a_x(z) = \mathbf{E}[\phi_x(X) \mid Z = z]$ and $m_y(z) = \mathbf{E}[\phi_y(Y) \mid Z = z]$. Two W terms are conditionally independent when their index sets $\{i, \pi(i)\}$ are disjoint.

Each set overlaps at most two others. Their conditional centered second moments are bounded by C^2 . Cauchy–Schwarz therefore gives

$$\mathbf{E} \left[\left\| \frac{1}{n} \sum_i (W_i - \mathbf{E}[W_i | \mathcal{F}_n]) \right\|^2 \middle| \mathcal{F}_n \right] \leq \frac{3C^2}{n}.$$

This centered average converges to zero in probability.

Continuity of m_y is not required. Universality of the product kernel implies that ϕ_z distinguishes points. Compactness and continuity therefore make d_z a metric inducing the topology of $\mathcal{Z}$. The feature spaces are separable. For any $\epsilon > 0$, choose a bounded continuous $g : \mathcal{Z} \rightarrow \mathcal{H}_y$ with $\mathbf{E} \|m_y(Z) - g(Z)\| < \epsilon$. Such an L^1 approximation follows by approximating the bounded measurable m_y by simple functions, then approximating their indicator functions by continuous scalar functions using regularity of P_Z . Since a permutation preserves empirical averages, for any $\eta > 0$,

$$\begin{aligned} A_n &:= \frac{1}{n} \sum_i \|m_y(Z_{\pi(i)}) - m_y(Z_i)\| \\ &\leq \frac{2}{n} \sum_i \|m_y(Z_i) - g(Z_i)\| \\ &\quad + \omega_g(\eta) + \frac{2\|g\|_\infty \delta_n}{\eta}, \end{aligned}$$

where ω_g is the modulus of continuity in d_z . For fixed g , the first term converges to a value below 2ϵ by the law of large numbers. Choose η small and let $n \rightarrow \infty$; then let $\epsilon \downarrow 0$. This proves $A_n \xrightarrow{P} 0$. The average conditional means thus differ from $n^{-1} \sum_i a_x(Z_i) \otimes m_y(Z_i) \otimes \phi_z(Z_i)$ by at most $C_x C_z A_n \xrightarrow{P} 0$. The bounded Hilbert-space law of large numbers gives convergence of this latter average to μ_Q . Consequently, the permuted empirical embedding converges to μ_Q , and the original empirical embedding converges to μ_P .

Set $V_n = n^{-2} \sum_{i,j} h(i,j)$ and $U_n = \{n(n-1)\}^{-1} \sum_{i \neq j} h(i,j)$. Then V_n is the squared distance between the two empirical embeddings. Since $|h(i,j)| \leq 4C^2$, $|V_n - U_n| \leq 8C^2/n$. If $r = n(n-1) - |\mathcal{J}| \leq 2n$ and S_E is the sum of the omitted off-diagonal terms, accounting for the changed denominator gives

$$U_n - T_n = \frac{S_E - rT_n}{n(n-1)}.$$

Both $|T_n| \leq 4C^2$ and $|S_E| \leq 4C^2 r$, so

$$|U_n - T_n| \leq \frac{8C^2 r}{n(n-1)} \leq \frac{16C^2}{n-1}.$$

This bound accounts for both deletion and renormalization, unlike the original argument. For $n > 3$, $|\mathcal{J}| \geq$

$n(n-3) > 0$. The two differences vanish, giving the claimed limit. Universality implies $\mu_P = \mu_Q$ exactly when $P = Q$, equivalently $X \perp\!\!\!\perp Y \mid Z$. This completes the proof.

Changes

2026-09-19. Sanghack Lee and Vasant Honavar, “Self-Discrepancy Conditional Independence Test,” *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence (UAI 2017)*. Original paper: <https://www.auai.org/uai2017/proceedings/papers/16.pdf>.

This version was prepared with AI assistance. **Author review pending.** It is an unofficial proposed correction, not a publisher-issued erratum; approval by all authors is not implied. It is based on the authors’ camera-ready source, including its original Appendices A and B. Appendix C supplies the expanded proof.

Location	Before → After	Assumptions and limits
Abstract; §4.1, Eq. (2)	Independent retained observations / unbiasedness language → index exclusion with residual dependence; no general finite-sample unbiasedness claim.	Conditioning on the learned permutation alone does not restore independence. Asymptotic convergence is a separate statement.
Theorem 1; App. C	Inherited convergence argument and an $8C^2/(n-1)$ deletion bound → an explicit conditional-dependence proof and a bound including renormalization, $16C^2/(n-1)$ in general.	i.i.d. observations on compact metric spaces; fixed bounded continuous kernels with universal product; a Z -only derangement with independent auxiliary randomness; mean RKHS distortion $\delta_n \rightarrow_P 0$. Conditional-mean continuity is not required. The conclusion concerns the raw statistic, not the level or consistency of the full test.
§4.2; App. B	Pair-count justification of the 1/2 rescaling and claims about the regression adjustment → empirical approximations, with the implemented adjustment specified.	Experiments support calibration in the evaluated settings. No general calibration theorem follows from pair counts, variance reduction, or Theorem 1. Adaptive distances and bandwidths need separate analysis.
§4.3; Alg. 4	Infinite penalties on retained pairs $\mathcal{J} \rightarrow$ penalties on excluded off-diagonal pairs $\mathcal{J}_\pi$; the algorithm now describes the finite $D_{\max}$ implementation penalty.	This is a soft penalty; reconstruction is not prohibited, and the penalty vanishes for $D = 0$. Hard exclusions can be infeasible.
§4.4	Strict empirical upper tail → $\hat{p} = (1 + \#\{N_r - \bar{N} \geq T\})/(b+1)$.	Includes ties and a finite Monte Carlo correction. It prevents the constant-data $p = 0$ defect, but does not validate the reference distribution.
§3.1; §5.2–5.4	Monotonic-power, equal-power, differing-null, kernel-robustness, and Blossom V complexity claims → qualified empirical comparisons and corrected attribution.	The CI null formulations are equivalent. Matching reference variances at $B = 2145$ on null data does not measure equal power. Larger B narrows KCIPT’s reference without removing variation across datasets.
§5.1; Figs. 1–6	Positive squared- Y term → negative term in the Hénon equation; a shared noise symbol → independent E_X, E_Y in the post-nonlinear null model.	The Hénon lag coefficient is 0.3 in the printed model and Python generator, 0.1 in archived MATLAB. Its value in the plotted datasets is unverified. Serial observations do not meet Theorem 1’s i.i.d. assumption.
Figure 2	CHSIC said to be excluded from the bottom row → included for dimensions 1–5; AUPC at dimensions 3–5 is below the displayed range.	Caption corrected against the retained plots and archived results; the plotted values and axis limits are unchanged.
Throughout	English, notation, references, and cross-references corrected.	Original authorship, publication year, and conference identity retained.

Experimental verification. Experiments were not rerun; reported experimental numerical values and figure assets are unchanged. Targeted code checks found defects in native matrix shuffling, matching-cost arithmetic, and strict-tail handling. Corrections can change outputs; their impact on the historical benchmark comparisons remains unverified. The unresolved Hénon model provenance is also noted beside the affected figures.

Citation. Cite the original 2017 paper. If this version was used, we recommend adding “Unofficial AI-corrected version dated 2026-09-19 consulted” and the URL of this version to the same reference’s note:
https://www.sanghacklee.me/ai-corrected/SDCIT_AI_corrected_2026-09-19.pdf.

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